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A non-technical, executive business impact compendium detailing **24 production systems and architectural case studies** engineered across the United States, United Kingdom, and Europe. Structured specifically for founders, CEOs, and non-developers to clearly explain **what painful problems existed**, **how they were solved with AI & automation**, and **what measurable business results were delivered**.

📁 Case Studies Hub: ahsantariq.ranbval.com/case-studies/ ↗ 🌐 Main Portfolio ↗ 📧 ahsantariq0724@gmail.com 🔗 LinkedIn Profile ↗

24	60% – 90%	$141M+	< 60s
IN-DEPTH CASE STUDIES	MANUAL EFFORT SLASHED	ACTIVE OPERATIONS MIRRORED	LEAD VOICE TURNAROUND

📖 How This Case Studies Guide Is Structured for Non-Developers:

Every project in this document follows a three-part business framework:

- **1. The Painful Problem Faced:** What manual bottleneck, lost revenue, or operational chaos the client suffered before.
- **2. The Smart Solution Built:** In plain English, how the software or AI system was designed to fix the root problem.
- **3. The Real-World Business Results:** Concrete hours saved, dollars cut, error rates eliminated, and direct links to live case studies.

IMPACT DEEP DIVES

Healthcare Enterprise Case Studies

Non-technical problem breakdowns and measurable operational results delivered for international business clients.

#01 Ambica International (Dovetail) — Healthcare Logistics & Enterprise ERP

LIVE PRODUCTION

Client / Organization: Ambica International · United States (Remote) **Timeline:** 2025 – 2026 **Domain:** Healthcare Enterprise · ERP & Live Telematics

⚠️ THE PAINFUL PROBLEM FACED

Hospitals and pharmacies were relying on paper clipboards, phone calls, and manual spreadsheet entries to track critical medicine shipments. Managers had zero live visibility into delivery trucks, purchase orders took days to manually key into NetSuite, and signing vendor contracts required printing physical paper.

💡 THE SMART SOLUTION BUILT

Built a 100% digital, autonomous operations portal hosted on Google Cloud Run. Connected live GPS sensors from 102 delivery vehicles, built a sub-second data bridge that syncs NetSuite inventory and invoices every 3 seconds, and integrated automated DocuSign workflows so executives can approve supplier contracts from their phones in under 60 seconds.

🚀 THE REAL-WORLD BUSINESS RESULTS DELIVERED

Eliminated lost shipments completely, automated tracking across 51,462 invoices and 35,935 purchase orders, reduced vendor contract turnaround from 3 days to under 1 minute, and gave executive leadership a live 3-second profit & loss dashboard with zero manual bookkeeping.

$141.3M	102 Trucks	< 60s	3-Sec
ACTIVE CARGO MIRRORED	LIVE CARTRACK GPS	DOCUSIGN APPROVALS	LIVE P&L TICKING

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/ambica-dovetail/ ↗

#02 Petroski Physiotherapy — Clinic Marketing & Patient Intake Governance

CLINICAL AUTOMATION

Client / Organization: Petroski Physio Clinic · Sports Rehabilitation (USA) **Timeline:** 2025 **Domain:** Healthcare Governance · Clinical Workflow

⚠️ THE PAINFUL PROBLEM FACED

Sports physical therapists were constantly being interrupted during patient therapy sessions by clinic marketers asking: 'Can we film this injured athlete for Instagram/TikTok?' This created HIPAA privacy risks, annoyed busy therapists, and resulted in missed marketing opportunities when therapists forgot to follow up.

💡 THE SMART SOLUTION BUILT

Designed an automated 'Two-Board' governance system. Clinical medical records are kept completely separated from marketing. Exactly 24 hours after a patient's evaluation, the therapist automatically receives a friendly Slack message asking: 'Can we film this athlete?' with simple 'Yes' and 'No' buttons. Clicking 'Yes' instantly and safely updates the marketing team's board.

🚀 THE REAL-WORLD BUSINESS RESULTS DELIVERED

Completely eliminated office interruptions during patient treatment, guaranteed 100% HIPAA medical privacy compliance with zero patient data leaks, and automatically archives patients once their physical therapy program concludes.

100%	24-Hour	1-Click	0 Interruptions
HIPAA PRIVACY SHIELD	AUTOMATED TRIGGER	SLACK APPROVAL	DURING PATIENT CARE

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/petroski-physio/ ↗

IMPACT DEEP DIVES

Global FinTech Case Studies

Non-technical problem breakdowns and measurable operational results delivered for international business clients.

#03 Volvexer — Global Multi-Ledger Accounting & Tax Reconciliation

ENTERPRISE PLATFORM

Client / Organization: Volvexer · Financial Services (USA, UK, Australia, Pakistan) **Timeline:** 2024 – Present **Domain:** Global FinTech · Multi-Jurisdiction Ledgers

⚠️ THE PAINFUL PROBLEM FACED

Multinational companies operating across the US, UK, Australia, and Pakistan were wasting hundreds of hours every month manually re-entering bank transactions, converting currencies, and matching invoices between different accounting software like QuickBooks, Xero, NetSuite, and Zoho Books.

💡 THE SMART SOLUTION BUILT

Co-founded and engineered an automated financial platform that synchronizes bank transactions, automatically converts international exchange rates, reads messy receipts with AI, and prepares tax reconciliation reports tailored to US (GAAP/IRS), UK (HMRC), and Australian tax rules without manual entry.

🚀 THE REAL-WORLD BUSINESS RESULTS DELIVERED

Reduced month-end financial closing time by 75%, eliminated duplicate payments, and established an encrypted, tamper-proof audit trail that satisfies international corporate tax audits.

75%	4 Countries	0 Re-typing	100%
FASTER MONTH-END CLOSE	TAX RULES INTEGRATED	MANUAL DATA ENTRY	AUDIT-READY TRAIL

Read Full Architectural Case Study: www.volvexer.com/ ↗

#04 Terry Properties — AI Accounting Automation & QuickBooks / Buildium Sync

PRODUCTION SYSTEM

Client / Organization: Terry Real Estate Management · United States (Remote) **Timeline:** 2024 – 2025 **Domain:** Real Estate FinTech · Invoice Reconciliation

⚠️ THE PAINFUL PROBLEM FACED

Property managers were spending 25+ hours every week manually opening hundreds of vendor bills from messy email attachments and WhatsApp photos, and manually typing invoice numbers, amounts, and dates into both QuickBooks Online and Buildium. Several maintenance contractors were accidentally paid twice.

💡 THE SMART SOLUTION BUILT

Engineered an AI accountant that automatically reads photos or PDF invoices, verifies whether the bill has already been paid or entered, extracts the contractor name and general ledger code, and syncs the expense record accurately across both QuickBooks and Buildium simultaneously.

🚀 THE REAL-WORLD BUSINESS RESULTS DELIVERED

Saved property management staff 25+ hours of tedious manual typing every single week, completely eradicated duplicate contractor payments, and gave property owners instant financial clarity on every repair bill.

25+ Hours	99.4%	0 Duplicate	2-Way
SAVED EVERY WEEK	AI INVOICE EXTRACTION	CONTRACTOR PAYMENTS	REAL-TIME SYNC

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/terry-accounting/ ↗

IMPACT DEEP DIVES

FinTech Sales Case Studies

Non-technical problem breakdowns and measurable operational results delivered for international business clients.

#05 Transact ULTD — Sub-Second Voice AI & Automated Merchant Lifecycle

VOICE AI · LIVE

Client / Organization: Transact ULTD · Payment Technology (USA) **Timeline:** 2025 – 2026 **Domain:** FinTech Sales · Real-Time Voice Specialist

⚠️ THE PAINFUL PROBLEM FACED

Credit card processing companies lose over 60% of interested business leads when sales calls go to voicemail or callers are placed on hold. Busy merchant owners hang up immediately if an automated phone system sounds robotic, stutters, or delays before replying.

💡 THE SMART SOLUTION BUILT

Engineered a lightning-fast AI phone specialist that calls incoming merchant leads in under 60 seconds. It talks smoothly like a knowledgeable consultant, explains fee savings on credit cards, parses merchant statements, and transfers live callers smoothly to human closing specialists when complex questions arise.

🚀 THE REAL-WORLD BUSINESS RESULTS DELIVERED

Cut inbound lead response time from hours to under 60 seconds, boosted sales appointment booking rates, and created a 24/7 automated qualification funnel that never puts a customer on hold.

< 60 Sec	< 600ms	0 Hold Times	5 Experts
LEAD RESPONSE TIME	CONVERSATIONAL DELAY	24/7 AVAILABILITY	LIVE TRANSFER MESH

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/transact-ultd/ ↗

#06 Django Voice Agent — Autonomous Phone AI & Meeting Action Summarizer

AUTONOMOUS AGENT

Client / Organization: Enterprise Sales Infrastructure · Commercial Deployment **Timeline:** 2025 – 2026 **Domain:** Autonomous Voice AI · Calendar Automation

⚠️ THE PAINFUL PROBLEM FACED

Sales teams lose valuable selling hours dialing cold phone numbers, emailing calendar links back and forth across different time zones, and taking manual meeting notes during client consultations.

💡 THE SMART SOLUTION BUILT

Built an autonomous phone assistant combining Twilio, Deepgram, and Claude. The agent speaks naturally, answers objections, checks team calendar availability, and reserves appointments on Google Meet or Zoom. When the meeting concludes, the AI automatically transcribes the recording and delivers concise next steps directly into Slack.

🚀 THE REAL-WORLD BUSINESS RESULTS DELIVERED

Eliminated back-and-forth calendar scheduling emails entirely, removed the burden of manual note-taking, and ensured sales managers receive instant action-item summaries within 2 minutes of every call.

100%	0 Note-Taking	< 2 Min	3 Platforms
HANDS-FREE BOOKING	AUTOMATED TRANSCRIPTS	SLACK ACTION DELIVERY	GOOGLE / ZOOM / TEAMS

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/django-voice-agent/ ↗

IMPACT DEEP DIVES

Consumer AI Case Studies

Non-technical problem breakdowns and measurable operational results delivered for international business clients.

#07 Qira AI — Real-Time Avatar Intelligence & Multilingual Digital Twins

CONSUMER PLATFORM

Client / Organization: Qira AI · United States & Global **Timeline:** 2024 – 2025 **Domain:** Consumer AI · Digital Twin Video Synthesis

⚠️ THE PAINFUL PROBLEM FACED

Content creators, influencers, and busy professionals want to stay connected with their fans and friends, but it is physically impossible to respond to thousands of video calls, chats, and direct messages around the clock.

💡 THE SMART SOLUTION BUILT

Built an entertainment platform where users create a speaking 3D digital clone from just one selfie and short voice sample. Friends can video call, text, and play games with the digital twin, which speaks authentically in the user's real voice, translates conversations across multiple languages, and interacts autonomously when the real user is asleep or busy.

🚀 THE REAL-WORLD BUSINESS RESULTS DELIVERED

Delivered lifelike real-time video chat with sub-800ms latency, enabled creators to scale their personal presence to thousands of fans simultaneously, and opened an entirely new category of interactive social entertainment.

1 Photo	< 800ms	Multi-Lang	24/7
AVATAR CREATION	REAL-TIME VIDEO SYNC	LIVE SPEECH TRANSLATION	AUTONOMOUS TWIN MODE

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/qira-ai/ ↗

#08 Ballyhass Adventure Park — Autonomous Booking Assistant & Availability

COMMERCIAL CHATBOT

Client / Organization: Ballyhass Outdoor Adventure · Ireland / Europe **Timeline:** 2024 – 2025 **Domain:** Hospitality & Leisure · Live Reservations

⚠️ THE PAINFUL PROBLEM FACED

Families planning weekend trips to a large outdoor lake park struggled to find open time slots across different sports like wakeboarding, ziplines, and aquaparks. The customer support phone line was constantly jammed with callers asking simple questions about ticket availability.

💡 THE SMART SOLUTION BUILT

Built a friendly booking assistant that understands everyday conversational questions like: 'Are there spots for 4 people next Friday afternoon at Mallow?' It checks live ticket availability in 0.4 seconds across all activity venues and provides direct buttons for visitors to book their preferred slot in one click.

🚀 THE REAL-WORLD BUSINESS RESULTS DELIVERED

Reduced customer support phone calls by over 50%, eliminated booking friction, and allowed thousands of visitors to reserve adventure passes effortlessly 24 hours a day.

0.4 Sec	1-Click	2 Lakes	24/7
AVAILABILITY LOOKUP	TICKET RESERVATION	MULTI-VENUE POLLING	AUTOMATED HELPDESK

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/ballyhass-booking/ ↗

IMPACT DEEP DIVES

Sales Enablement Case Studies

Non-technical problem breakdowns and measurable operational results delivered for international business clients.

#09 Prospect Intelligence Engine — Sales AI with Google Context Caching

ENTERPRISE AI

Client / Organization: Enterprise Sales Intelligence · Contract Engagement **Timeline:** 2025 – 2026 **Domain:** Sales Enablement · LLM Server Caching

⚠️ THE PAINFUL PROBLEM FACED

Sales representatives wasted 10 to 15 minutes before every call digging through old call recordings, LinkedIn profiles, and company PDFs to answer buyer questions. Asking conventional AI to re-read 100,000 words on every question was too slow (10+ seconds) and cost a small fortune in API fees.

💡 THE SMART SOLUTION BUILT

Engineered an AI sales assistant utilizing Google Gemini's advanced server caching. The system loads the client's complete historical profile into server memory once; future buyer questions are answered in less than 1 second while bypassing 90% of the normal AI computing charges.

🚀 THE REAL-WORLD BUSINESS RESULTS DELIVERED

Slashed monthly AI infrastructure bills by over 90%, cut answer latency from 10 seconds to under 800 milliseconds, and equipped sales reps with instant, accurate prospect knowledge during high-stakes sales conversations.

90%	< 1 Sec	100k Words	10 Min -> 0
AI BILL SLASHED	QUERY RESPONSE TIME	INSTANT CONTEXT MEMORY	PRE-CALL PREP TIME

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/prospect-intelligence/ ↗

#10 hddisenos — Self-Hosted Automation Cloud & Multi-CRM Synchronization

SELF-HOSTED CLOUD

Client / Organization: hddisenos Creative & Industrial Agency · International **Timeline:** 2025 **Domain:** Private Cloud DevOps · Multi-CRM Workflows

⚠️ THE PAINFUL PROBLEM FACED

Commercial automation platforms (like Zapier or Make) charge for every single click, workflow trigger, and data transfer. As the agency scaled, software bills ballooned into thousands of dollars each month, while team members still had to manually sync leads between HubSpot and Zoho CRM.

💡 THE SMART SOLUTION BUILT

Built a private, self-hosted automation cloud on dedicated servers. It powers 57 business operations—including turning customer pet photos into AI watercolor art, automatically posting car inventory to Facebook Marketplace for auto dealerships, and synchronizing customer records across HubSpot and Zoho with zero third-party software subscriptions.

🚀 THE REAL-WORLD BUSINESS RESULTS DELIVERED

Cut manual administrative office work by 60%, eliminated thousands of dollars in monthly SaaS subscription fees forever, and delivered 100% data ownership and privacy on dedicated servers.

60%	$0 / Month	57 Automated	2 Major CRMs
MANUAL WORK CUT	RECURRING TOOL FEES	BUSINESS WORKFLOWS	HUBSPOT & ZOHO SYNC

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/hddisenos-automation/ ↗

IMPACT DEEP DIVES

Film Production AI Case Studies

Non-technical problem breakdowns and measurable operational results delivered for international business clients.

#11 CoreImmersive — Hollywood Call Sheet Parsing & Set Telematics

MEDIA PRODUCTION AI

Client / Organization: CoreImmersive Productions · Hollywood / Los Angeles (Remote) **Timeline:** 2024 – 2025 **Domain:** Film Production AI · Logistics & Telematics

⚠️ THE PAINFUL PROBLEM FACED

Filming a movie or television series is high-stakes logistical chaos: hundreds of actors, camera crews, and catering trucks operate off complicated 20-page PDF 'Call Sheets' emailed late at night. Production assistants spent hours manually copying schedules, missing rainstorms, and triggering expensive union meal penalty fines.

💡 THE SMART SOLUTION BUILT

Built an intelligent production management platform. The AI scans dense multi-page schedule PDFs in seconds, organizes arrival times and shooting locations for every actor, tracks union meal break countdowns, and monitors weather radar to warn directors 90 minutes before rain hits outdoor filming sets.

🚀 THE REAL-WORLD BUSINESS RESULTS DELIVERED

Eliminated expensive union delay penalties, automated call sheet distribution to cast and crew, protected filming equipment from sudden weather damage, and gave Hollywood directors a live progress dashboard on set.

99.2%	90 Min	$0 Union	100% Real-Time
PDF CALL SHEET ACCURACY	ADVANCE RAIN WARNING	MEAL PENALTY FINES	DIRECTOR DASHBOARDS

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/coreimmersive-film/ ↗

#12 Sean Z & Associates — Multilingual AI Legal Intake Architecture

LEGAL AI ASSISTANT

Client / Organization: Sean Z & Associates · Law Practice (USA) **Timeline:** 2024 **Domain:** Legal Technology · Multilingual Case Prep

⚠️ THE PAINFUL PROBLEM FACED

Immigration lawyers spend endless hours conducting repetitive initial screening interviews in English and Spanish over the phone. Potential clients who contacted the firm after business hours were lost to competitor law firms before staff could respond.

💡 THE SMART SOLUTION BUILT

Built an intelligent legal intake assistant accessible via WhatsApp and phone. The AI interviews potential clients in their native language, asks relevant questions about family history and visa requirements, structures the facts into a clean case dossier, and alerts attorneys immediately while ensuring all data remains confidential.

🚀 THE REAL-WORLD BUSINESS RESULTS DELIVERED

Cut case file preparation time by 70%, captured after-hours client leads that were previously lost, and freed attorneys from repetitive paperwork so they could focus on courtroom advocacy.

2 Languages	100%	70% Faster	24/7
ENGLISH & SPANISH	ATTORNEY-CLIENT PRIVILEGE	CASE FILE PREPARATION	CLIENT INTAKE ON WHATSAPP

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/seanz-law-firm/ ↗

IMPACT DEEP DIVES

DevOps & Cloud Computing Case Studies

Non-technical problem breakdowns and measurable operational results delivered for international business clients.

#13 Jericho AI — Autonomous GenAI Ops & Multi-Cloud Cost Orchestrator

MULTI-CLOUD PLATFORM

Client / Organization: Jericho AI · Florida, United States **Timeline:** 2024 – 2025 **Domain:** DevOps & Cloud Computing · Multi-Cloud FinOps

⚠️ THE PAINFUL PROBLEM FACED

Enterprises run software across multiple cloud providers (AWS, Azure, Google Cloud, IBM, and Oracle). Because every platform has different dashboards and billing rules, companies waste thousands of dollars every month on idle, over-provisioned servers and cloud bandwidth.

💡 THE SMART SOLUTION BUILT

Engineered a unified multi-cloud orchestration platform powered by autonomous AI agents. The agents continuously monitor server loads across all five cloud providers, identify unused computing power, and automatically restructure workloads in real time to stop waste without causing service interruptions.

🚀 THE REAL-WORLD BUSINESS RESULTS DELIVERED

Cut unnecessary cloud spending by over 35%, automated infrastructure optimization across multi-cloud environments, and gave engineering leadership a single unified dashboard to govern all company cloud assets.

5 Clouds	35%+	0 Downtime	24/7 AI
AWS, AZURE, GCP, IBM, OCI	CLOUD WASTE REDUCTION	AUTOMATED SCALING	AUTONOMOUS SERVER SWARM

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/jericho-ai-multicloud/ ↗

#14 InstaMailAI — Scalable Vector Database & Semantic RAG Email Engine

VECTOR SEARCH RAG

Client / Organization: InstaMailAI · United States (Remote) **Timeline:** 2025 **Domain:** Vector Databases · Semantic Search & RAG

⚠️ THE PAINFUL PROBLEM FACED

Traditional email search requires users to remember the exact keyword used months ago. When corporate teams handle thousands of customer emails, finding past context or drafting accurate, customized replies takes minutes per email, slowing down customer service.

💡 THE SMART SOLUTION BUILT

Built a scalable vector search engine using MongoDB Atlas Vector Search and OpenAI embeddings. The system indexes millions of email messages into mathematical representations of meaning; users can find past discussions instantly using everyday concept searches, and the AI drafts context-aware responses automatically.

🚀 THE REAL-WORLD BUSINESS RESULTS DELIVERED

Accelerated email drafting speed by 80%, allowed teams to find past client agreements in less than 50 milliseconds, and eliminated customer service delays across high-volume email workflows.

< 50ms	80%	0 Hallucinations	Millions
VECTOR SEARCH SPEED	FASTER EMAIL DRAFTING	CONTEXT-GROUNDED REPLIES	INDEXED EMAIL THREADS

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/instamail-ai/ ↗

IMPACT DEEP DIVES

AI SaaS Architecture Case Studies

Non-technical problem breakdowns and measurable operational results delivered for international business clients.

#15 2456.ai — Autonomous AI Tool Discovery & Scalable Microservices

SAAS PLATFORM

Client / Organization: 2456.ai · United States (Remote) **Timeline:** 2023 – 2024 **Domain:** AI SaaS Architecture · Microservices

⚠️ THE PAINFUL PROBLEM FACED

With thousands of new AI tools launching every week, businesses and professionals struggle to identify which AI model actually solves their specific problem, leading to wasted subscription fees and confusion.

💡 THE SMART SOLUTION BUILT

Built an intelligent AI platform that automatically catalogs AI tools, analyzes their performance capabilities, and routes user questions to the optimal tool with sub-second response times, integrated vector recommendations, and automated tiered Stripe subscription billing.

🚀 THE REAL-WORLD BUSINESS RESULTS DELIVERED

Created an effortless AI discovery hub that routes queries in under 1 second, automated recurring customer billing, and handled high concurrent user search traffic without server slowdowns.

Thousands	< 1 Sec	100%	Zero-Lag
AI TOOLS INDEXED	MODEL ROUTING LATENCY	AUTOMATED STRIPE BILLING	CONCURRENT SEARCH

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/2456-ai-platform/ ↗

#16 QPharma Global — Cloud Data Lake & Cold-Chain Telemetry Pipeline

ENTERPRISE DATA LAKE

Client / Organization: QPharma Global · United States (Remote) **Timeline:** 2022 **Domain:** Big Data Engineering · Cloud Data Lake

⚠️ THE PAINFUL PROBLEM FACED

Pharmaceutical companies must guarantee that temperature-sensitive vaccines and medicines never spoil during international transportation. Manually processing gigabytes of sensor temperature logs and pharmacy sales spreadsheets took days, delaying critical regulatory compliance reports.

💡 THE SMART SOLUTION BUILT

Built a serverless cloud data lake on Amazon Web Services (AWS Glue, S3, and Lambda). The pipeline automatically ingests, cleans, and partitions millions of shipment logs, alerting quality managers instantly if medicine temperature limits were breached.

🚀 THE REAL-WORLD BUSINESS RESULTS DELIVERED

Cut data processing and compliance reporting time by 70%, prevented damaged pharmaceutical batches from reaching pharmacy shelves, and delivered sub-second audit queries for federal regulators.

70%	Millions	100%	Sub-Second
FASTER DATA REPORTING	RECORDS CLEANSED	COLD-CHAIN INTEGRITY	ATHENA SQL QUERIES

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/qpharma-data-platform/ ↗

IMPACT DEEP DIVES

Cryptographic Security Case Studies

Non-technical problem breakdowns and measurable operational results delivered for international business clients.

#17 Ranbval Core & Secret Manager — Zero-Knowledge Cryptographic Vault

NLNET GRANT AWARD

Client / Organization: Ranbval Security Platform · Global Enterprise SaaS **Timeline:** 2025 – Present **Domain:** Cryptographic Security · Zero-Trust Architecture

⚠️ THE PAINFUL PROBLEM FACED

Over 80% of corporate data breaches occur because software engineers leave database passwords, customer encryption keys, and API tokens stored in plain text configuration files on servers or in GitHub repositories.

💡 THE SMART SOLUTION BUILT

Built an ultra-secure digital vault that encrypts confidential credentials using military-grade cryptography (AES-256 and ChaCha20). Keys are verified offline in less than 2 milliseconds, and if unauthorized tampering is detected, the system shuts down access instantly. Awarded an EU NLnet Foundation research grant.

🚀 THE REAL-WORLD BUSINESS RESULTS DELIVERED

Protected production servers from credential theft, published open-source verification libraries on PyPI and npm with over 11,500 global downloads, and achieved verified sub-2ms decryption latency.

< 2ms	11,500+	NLnet EU	0 Leaks
KEY DECRYPTION SPEED	SDK DOWNLOADS GLOBALLY	RESEARCH GRANT WINNER	TAMPER-PROOF KILLSWITCH

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/ranbval-core/ ↗

#18 Ranbval Social Super-System — 16-Pillar Collaborative Platform

PROPRIETARY PLATFORM

Client / Organization: Ranbval Platform · Lead Systems Architect **Timeline:** 2024 – Present **Domain:** Unified SaaS · Single Secure Root

⚠️ THE PAINFUL PROBLEM FACED

Modern business teams waste productive hours jumping between 10 separate apps for team messaging, task management, file storage, video meetings, and AI helpers—paying separate subscriptions for each and scattering data everywhere.

💡 THE SMART SOLUTION BUILT

Architected an all-in-one collaborative ecosystem operating on a single secure web address (ranbval.com). Combines instant team chat, automated AI assistants, encrypted file vaults, interactive project boards, and group calls under one login protected by vault-first security.

🚀 THE REAL-WORLD BUSINESS RESULTS DELIVERED

Eliminated the chaos and cost of subscribing to 10 disconnected software tools, unified team communication in under 50 milliseconds, and delivered total data privacy under a single secure umbrella.

16 Apps	< 50ms	100%	10 Apps
UNIFIED UNDER 1 LOGIN	REAL-TIME CHAT LATENCY	SINGLE SECURE WEB ROOT	REPLACED BY RANBVAL

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/ranbval-social/ ↗

IMPACT DEEP DIVES

Omnichannel CRM Case Studies

Non-technical problem breakdowns and measurable operational results delivered for international business clients.

#19 Ranbval Conversations — Centralized Client Memory & Omnichannel Hub

PROPRIETARY PLATFORM

Client / Organization: Ranbval Platform · Lead Systems Architect **Timeline:** 2024 – Present **Domain:** Omnichannel CRM · Customer Memory

⚠️ THE PAINFUL PROBLEM FACED

When customers contact a business across different channels (email on Monday, WhatsApp on Wednesday, LinkedIn on Friday), support reps lose track of past conversations, frustrating clients who are forced to repeat themselves.

💡 THE SMART SOLUTION BUILT

Built a unified client communication hub where incoming messages from WhatsApp, email, LinkedIn, and websites flow into a single smart inbox. The AI remembers every customer's previous requests and purchases, instantly briefing the team before they answer.

🚀 THE REAL-WORLD BUSINESS RESULTS DELIVERED

Ensured no customer inquiry is ever forgotten, eliminated customer frustration from repeating past issues, and boosted customer satisfaction scores across multi-channel sales teams.

1 Inbox	100%	0 Repetitions	Instant
WHATSAPP, EMAIL, LINKEDIN	HISTORICAL CONTEXT MEMORY	NO REPEATED QUESTIONS	UNIFIED CLIENT PROFILE

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/ranbval-conversations/ ↗

#20 Industrial Defect Localization — Zero-Shot vs Few-Shot Vision Benchmark

ACADEMIC RESEARCH

Client / Organization: Peer-Reviewed Deep Learning Research · Sole Author: Ahsan Tariq **Timeline:** 2025 – 2026 **Domain:** Computer Vision · Industrial Quality Control

⚠️ THE PAINFUL PROBLEM FACED

Factory production lines lose millions when computer vision cameras fail to spot microscopic defects in manufactured products (cracked metal tiles, torn fabrics, loose screws), or when traditional AI models demand thousands of sample photos of defects before they can learn to spot them.

💡 THE SMART SOLUTION BUILT

Conducted comprehensive peer-reviewed computer vision research evaluating three modern AI paradigms across 15 industrial manufacturing categories (5,354 high-resolution images). Proved which visual foundation models accurately pinpoint microscopic manufacturing flaws with zero prior training photos.

🚀 THE REAL-WORLD BUSINESS RESULTS DELIVERED

Demonstrated how manufacturing plants can deploy visual AI quality control immediately without wasting months collecting rare defect samples, establishing empirical benchmarks for automated zero-defect manufacturing.

15 Categories	5,354	0 Prior Defect	Published
INDUSTRIAL OBJECTS & TEXTURES	HIGH-RES FACTORY IMAGES	PHOTOS REQUIRED TO TEST	PEER-REVIEWED STUDY

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/mvtec-defect-benchmark/ ↗

IMPACT DEEP DIVES

Decentralized AI Case Studies

Non-technical problem breakdowns and measurable operational results delivered for international business clients.

#21 Robust Federated Learning — Defense Against Byzantine Model Poisoning

ACADEMIC RESEARCH

Client / Organization: Peer-Reviewed Research · Sole Author: Ahsan Tariq **Timeline:** 2025 – 2026 **Domain:** Decentralized AI · Cybersecurity Defense

⚠️ THE PAINFUL PROBLEM FACED

Hospitals or fleets of autonomous vehicles need to train shared AI models together without exposing sensitive patient medical records or vehicle GPS coordinates to central servers. However, if a rogue or hacked device submits corrupted data, it can poison and crash the shared AI brain.

💡 THE SMART SOLUTION BUILT

Engineered a mathematical 'Trust Gate' algorithm that inspects AI model updates from every participant before accepting them into the global network. It detects subtle mathematical misalignments, instantly blocking and rejecting malicious updates without needing to look at private raw data.

🚀 THE REAL-WORLD BUSINESS RESULTS DELIVERED

Maintained 99.1% AI model accuracy even under coordinated cyber sabotage where standard federated learning failed completely, paving the way for safe, privacy-preserving AI collaboration across hospitals and connected vehicles.

99.1%	0 Private Data	Byzantine	Published
ATTACK REJECTION RATE	TRANSFERRED OR LEAKED	POISONING SHIELD	PEER-REVIEWED STUDY

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/robust-federated-perception/ ↗

#22 High-Throughput Network Intrusion Detection — ANOVA F-Test NIDS

ACADEMIC RESEARCH

Client / Organization: Peer-Reviewed Research · Sole Author: Ahsan Tariq **Timeline:** 2025 – 2026 **Domain:** Cybersecurity & Machine Learning · Real-Time Defense

⚠️ THE PAINFUL PROBLEM FACED

Modern enterprise networks face thousands of sophisticated hacking attempts, malware intrusions, and denial-of-service attacks every hour. Traditional AI firewalls get overwhelmed trying to analyze dozens of complex network attributes for every single packet, causing network lag or letting stealthy attacks slip through.

💡 THE SMART SOLUTION BUILT

Engineered a fast, lightweight cybersecurity defense pipeline using ANOVA statistical mathematics. It strips away 70% of network data clutter, focusing exclusively on the 15 most critical threat signals to identify cyber attacks with 98.67% accuracy in less than a millisecond.

🚀 THE REAL-WORLD BUSINESS RESULTS DELIVERED

Achieved state-of-the-art 98.67% threat detection accuracy on the modern UNSW-NB15 cybersecurity dataset, outperforming slower and more expensive deep learning firewalls while maintaining lightning-fast network speeds.

98.67%	69.4%	< 1ms	82,332
CYBER THREAT ACCURACY	NETWORK CLUTTER REMOVED	THREAT DETECTION LATENCY	NETWORK FLOW RECORDS TESTED

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/unsw-intrusion-detection/ ↗

IMPACT DEEP DIVES

Field Robotics Case Studies

Non-technical problem breakdowns and measurable operational results delivered for international business clients.

#23 Clearpath Husky A200 UGV — Autonomous 4x4 SLAM & Dijkstra Navigation

FIELD ROBOTICS

Client / Organization: Field Robotics & Autonomous Systems · Clearpath Robotics **Timeline:** 2025 – 2026 **Domain:** Field Robotics · Autonomous Navigation & SLAM

⚠️ THE PAINFUL PROBLEM FACED

Operating heavy-duty industrial vehicles in hazardous, unmapped environments (like mining tunnels, search-and-rescue disaster sites, or construction areas) is dangerous for human drivers, while remote-controlled vehicles lose signal and crash into obstacles.

💡 THE SMART SOLUTION BUILT

Engineered a complete autonomous navigation brain on a rugged Clearpath Husky 4x4 outdoor robot in ROS and Gazebo. The robot scans its surroundings with laser sensors, builds an accurate digital 2D floorplan in real time, and calculates the safest obstacle-free driving route autonomously.

🚀 THE REAL-WORLD BUSINESS RESULTS DELIVERED

Enabled unmanned rugged ground robots to navigate dangerous, unmapped territory safely without human drivers, accurately building 20mm resolution terrain maps while navigating around sudden obstacles.

20mm	100%	0 Human	1.07m
LASER MAP RESOLUTION	AUTONOMOUS OBSTACLE AVOIDANCE	REMOTE CONTROL REQUIRED	CALIBRATED WHEELBASE

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/husky-slam-navigation/ ↗

#24 Hector Quadrotor UAV & Smart Wheelchair — Flight Dynamics & Autonav

AERIAL & GROUND ROBOTICS

Client / Organization: Autonomous Drone & Hospital Mobility · ROS & Gazebo 3D **Timeline:** 2025 – 2026 **Domain:** Aerial Drone & Assistive Mobility · 3D Flight Control

⚠️ THE PAINFUL PROBLEM FACED

1) Drones in tight indoor spaces drift, shake, and crash due to air turbulence and wall reflections without GPS. 2) Wheelchair-bound hospital patients struggle to navigate narrow clinic corridors, constantly scraping door frames and walls.

💡 THE SMART SOLUTION BUILT

Engineered two high-precision autonomous navigation systems: 1) An autonomous quadrotor drone equipped with a 100 Hz cascade PID flight controller that takes off, hovers steadily in mid-air with sub-5cm altitude error, and flies through 3D waypoints. 2) A self-driving smart wheelchair that uses laser scanners to center itself through narrow hospital doorways with zero collisions.

🚀 THE REAL-WORLD BUSINESS RESULTS DELIVERED

Delivered sub-5cm flight stability for autonomous indoor drones without GPS, and gave hospital patients a crash-proof autonomous wheelchair that navigates narrow clinic hallways safely.

< 5cm	100 Hz	100%	0 Collisions
DRONE ALTITUDE HOVER ERROR	PID FLIGHT FEEDBACK	DOORWAY PASSAGE SAFETY	IN HOSPITAL CORRIDORS

Read Full Architectural Case Study: ahsantariq.ranbval.com/case-studies/hector-quadrotor-uav/ ↗

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Whether you need an autonomous phone assistant to capture customer leads, an automated accounting system to eliminate double-payments, or a multi-cloud architecture that slashes your server bills—I engineer production systems designed to pay for themselves through immediate operational savings.

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